

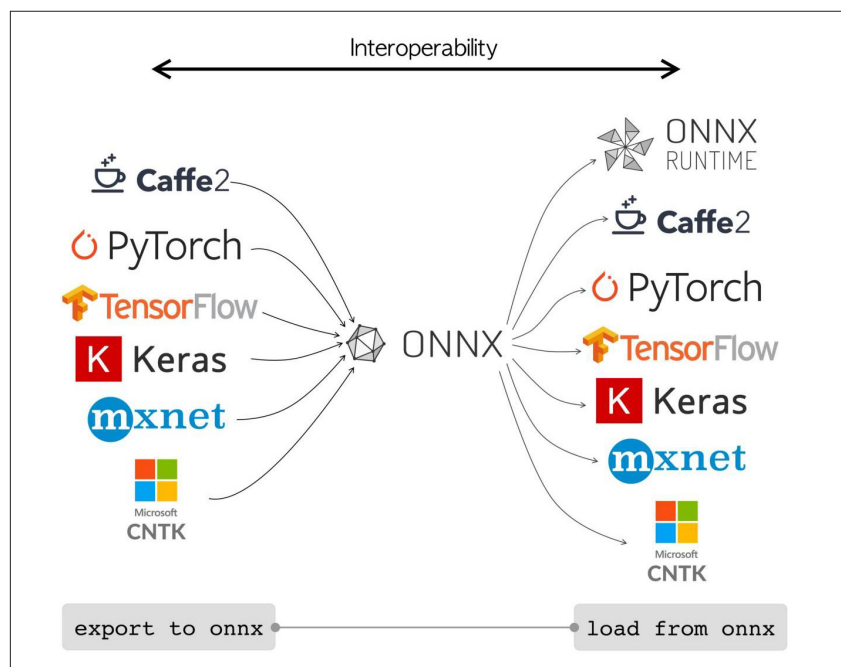


Figure 1: Interoperability through ONNX across various machine learning ecosystems

Table 1: YOLOv8 versions with their corresponding specifications

Model	Size (pixels)	mAP ^{val} 50-95	Speed (ms) CPU ONNX	Speed (ms) A100 TensorRT	params (M)	FLOPs (B)
YOLOv8n	640	37.3	80.4	0.99	3.2	8.7
YOLOv8s	640	44.9	128.4	1.20	11.2	28.6
YOLOv8m	640	50.2	234.7	1.83	25.9	78.9
YOLOv8l	640	52.9	375.2	2.39	43.7	165.2
YOLOv8x	640	53.9	479.1	3.53	68.2	257.8

These models are most frequently seen in data format (.pth or HD5) files. They should frequently be portable so that you can use them in contexts other than the ones where you trained the model. This is where ONNX is helpful. On a broad scale, ONNX is made to provide framework interoperability.

The process of plastic waste detection

ONNX simplifies the process of converting models between different frameworks. This means developers can more easily switch between libraries without having to start from scratch.

The ONNX runtime engine allows for high-performance inference across platforms.

PyTorch, TensorFlow, MXNet, and Caffe are just a few that have recently gained a lot of attention among the many outstanding machine learning libraries available in a variety of

languages. Figure 1 shows a few such machine learning ecosystems.

The output of an object detector is a set of bounding boxes that enclose the objects in the image, along with class labels and confidence scores for each box. YOLOv8 pretrained Detect models are shown here. Detect, Segment and Pose models are pretrained on the COCO data set, while Classify models are pretrained on the ImageNet data set. Table 1 lists the various YOLOv8 versions and their relevant parameters.

- **mAP^{val}** values are for single-model single-scale on the COCO val2017 data set. Reproduce by `yolo val detect data=coco.yaml device=0`
- **Speed** is averaged over COCO val images using an Amazon EC2 P4d instance. Reproduce by `yolo val detect data=coco128.yaml batch=1 device=0|cpu`

In the proposed set up, we have exported the ocean waste object detection data set containing 5136 images that is implemented using Google Colab. The resulting ONNX format can detect plastic under water using Ultralytics YOLOv8.

Step 1: Installing the relevant libraries

Installing libraries is the basic step for any program. So install Ultralytics for YOLOv8 and Roboflow to directly take images and annotations from the platform itself. Figure 2 shows the libraries installed.

```
installing libraries

!pip install ultralytics
!pip install roboflow
```

Figure 2: The libraries installed for the proposed task

```
checking proper installations for ultralytics

ultralytics.checks()
```

Figure 3: Checking for the proper installation of the libraries installed for the proposed task